JOB SEQUENCING WITH DEADLINE

J_1 J_2 J_3 J_4 J_5

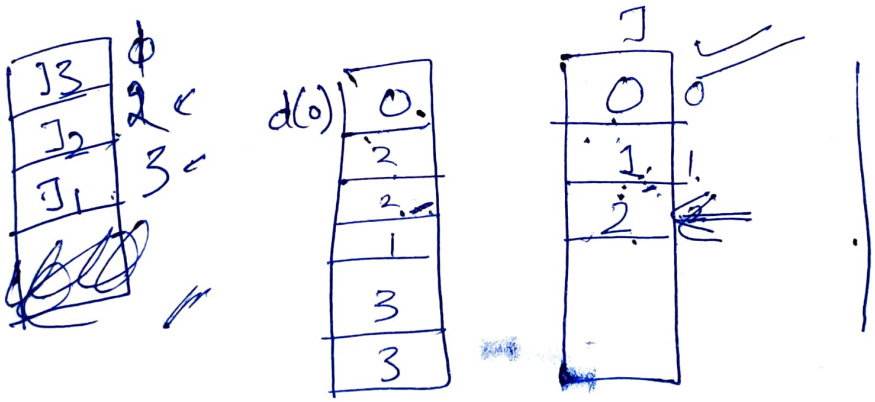
P = { 20, 15, 10, 5, 1 }

d = { 3, 3, 3, 3, 3 }.

{ J_1, J_2 J_4 }

{ J_2, J_1, J_4 }

1 2 3 4 5



k = 1

r = 1

i = 2

1

2 > 2

P = { _____ }

d = { 2, 1, 1, 3, 3 }.

```
Algorithm JS(d, J, n)
{
// Job must be arranged such that P_i ≥ P_{i+1}
d[0] = J[0] = 0;
J[1] = 1        // Assign first job
K = 1           // No. of jobs taken
for i=2 to n do
{
   r=K
   while( d[J[r]] > d[i] and (d[J[r]] ≠ r) // whether shifting is reqd. and feasible
   {
      r = r-1;
   }
   if (d[J[r]] ≤ d[i]) and (d[i] > r) // feasibility of inserting new job
   {
      for q = K to r+1 step -1 do J[q+1] = J[q];
                  // shift existing job downwards.
      J[r+1] = i;
      K = K+1;
   }
}
return K;
}
```

K = 1

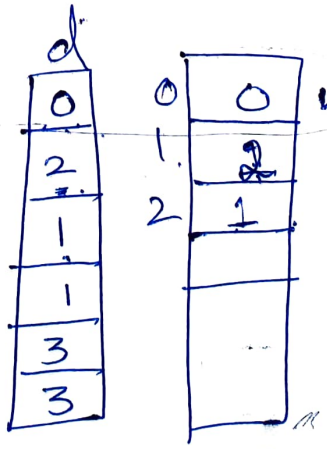
i = 2

r = 1

(2 > 1 && 2 ≠ 1)

r = 0

0 ≤ 1 and 1 > 0



K = 2

```
Algo minPlatform(arr, dep[], n){
    Plat=0, min=1
while(i<n && j<n){
    if(arr[i]<=dep[j])
    {  Plat++
       i++
    }
    else if(arr[i]>deep[j])
    {
       Plat--;
       j++
    }
    if(Plat > min)
       min = plat
}
return min
}
```

Time Complexity = $2n = O(n)$ (Sorted given)

$= 2n\log n + 2n = O(n\log n)$ ~~sort~~

Sweep line Algorithm

↳ HINT Convert (arr, dep) to x, y coordinates

& apply sweep line algo.

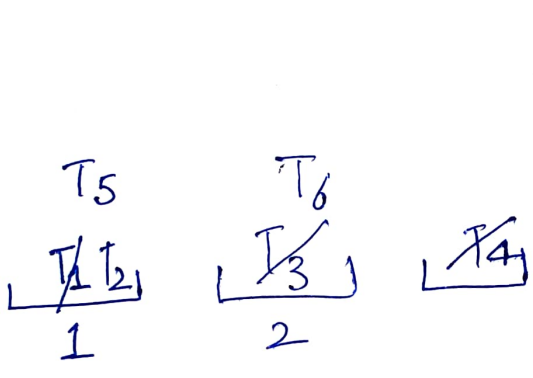
Minimum Platform Problem

Given

↳ Trains arrival and departure time for a station

↳ Minimize the ~~not~~ number of platforms reqd. to accomodate all trains.

	1	2	3	4	5	6
arr[] = [	0900	0945	0955	1100	1500	1800]
dep[] = [	0920	1200	1130	1150	1900	2000]



$d_i < a_j$

~~$a_j < d_i$~~

$d_j < a_i$

	T_1	T_2	T_3	T_4	T_5	T_6	
[Platfm 1		3	3	3	2	2]	max

max = 1

for (i = 1 to n).

for (j = 1 to n)

if (i ≠ j).

(intersection)

cap (max).

arr[] = [0900 0945 0955 1100 1500 1800]

dep[] = [0920 1130 1150 1200 1900 2000]

[0→~~1~~ 0 1 2 ~~3~~ ~~2~~ ~~1~~ 0 1 ~~2~~ 1]

3